Intro to Hands-on Tutorial 2

Interfacing Flux Scaling Approaches Across Space and Time

Presenters: Stefan Metzger and Sam Bower, AtmoFacts

Funding Acknowledgements: NSF Award 2532339 "*Conference: Remote Sensing and Fluxes for Real-world Impact: Integration with Advanced Techniques*"; Google Award "*BenchFlux: Scale-aware Benchmarks for Nature-based Climate Solutions Using Flux Measurements and Earth Observations*"; NSF Award 2313772 "*Closing the energy balance gap at scale*"

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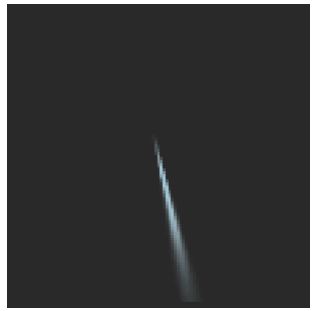
Approaches for Combining Fluxes and Remote Sensing

Selecting the right tool for the job

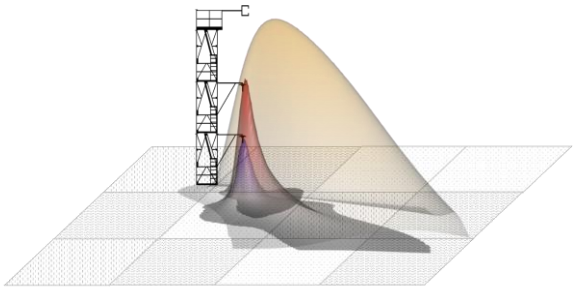
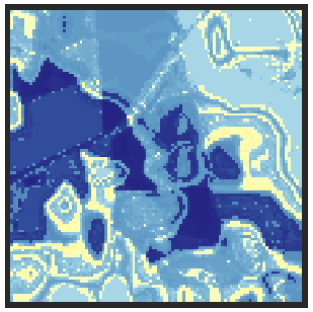
If you need…	Best Fit	Tutorial
Self-contained measurement	Classical Eddy Covariance	Tutorial 2 “AmeriFlux”
Site-specific accountability	Spatialized Eddy Covariance	Tutorial 2 “SpatialEddy”
Rapid regional extrapolation	Multi-Time, Multi-Site	Tutorial 2 “X-BASE”
Mechanistic hypothesis testing	Knowledge-Guided Machine Learning	Tutorial 2 “Eco2Bench”
Broad global coverage	Foundation Models	Tutorial 3

Let’s take a tour of the different approaches!

Self-Contained: Classical Eddy Covariance

	Eddy Covariance	Connectors	Remote Sensing
Input *Evidence*	Time-aggregate: 30-min+; single-site		
Intersect *Training Data*			
Relate *Training*			
Output *Inference*			

Sensor Fusion: Spatialized Eddy Covariance

	Eddy Covariance	Connectors	Remote Sensing
Input *Evidence*	Measurement: 10 Hz; single-time; single-site		Measurement: Indices
Intersect *Training Data*		4D perspective-matching	
Relate *Training*		First-principles boundary conditions	
Output *Inference*			

Metzger (2018)

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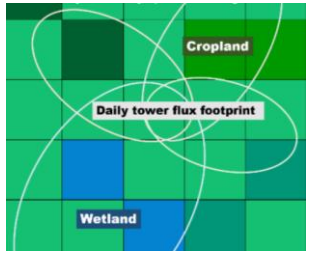
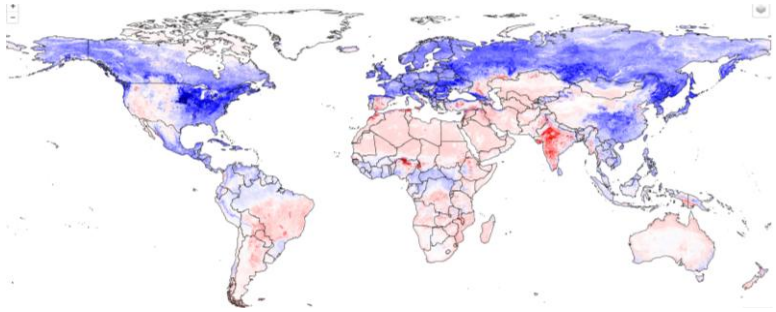
Data Integration: Multi-Time, Multi-Site

	Eddy Covariance	Connectors	Remote Sensing
Input *Evidence*	Time-space aggregate: 30-min+; multi-time; multi-site		Measurement (Indices)
Intersect *Training Data*		1D … 3D perspective-matching	
Relate *Training*		"More is better"	
Output *Inference*			

Kong et al. (2022); Nelson et al. (2023)

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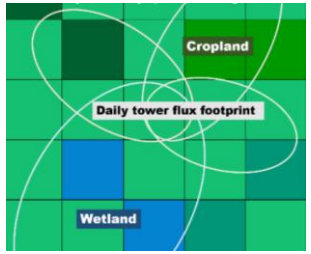
Data Integration: Knowledge-Guided Machine Learning

Eddy Covariance | **Connectors** | **Remote Sensing**

Input

Evidence

Time-space aggregate: 30-min+; multi-time; multi-site



Measurement (Indices)

Intersect

Training Data

1D … 3D perspective-matching

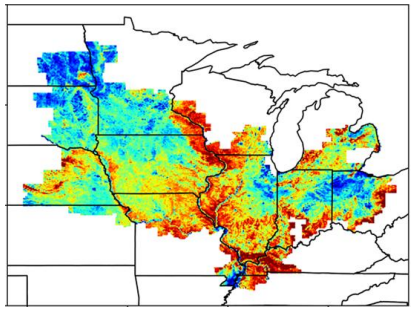
Relate

Training

Flux ~ RS process augmentation

Output

Inference



Kong et al. (2022); Liu et al. (2024)

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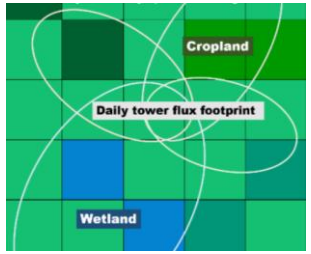
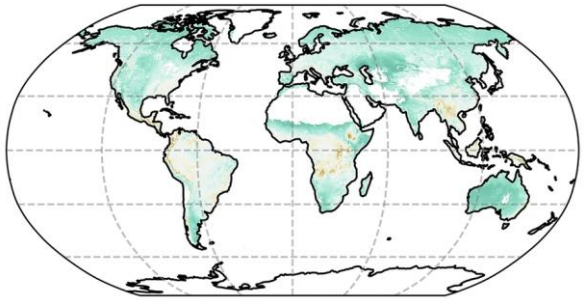
Data Integration: Foundation Models

	Eddy Covariance	Connectors	Remote Sensing
Input *Evidence*	Time-space aggregate: 30-min+; multi-time; multi-site		Process aggregate (embeddings)
Intersect *Training Data*		1D … 3D perspective-matching	
Relate *Training*		Process abstraction (embeddings)	
Output *Inference*			

Kong et al. (2022); Yuan et al. (2025)

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Differentiation and Complementarity among Approaches

Completeness & abstraction

Specificity & transparency

Feature	Classical EC	Spatialized EC	Multi-Time, Multi-Site	Knowledge-Guided	Foundation Models
Causal Transparency	✓	✓	◐	◐	-
Captures >95% emissions / removals (e.g., LCFS*)	-	✓	-	✓	-
Fully Localized in Space × Time	-	✓	◐	◐	◐
Uses Surrogate Evidence	-	-	✓	✓	✓
Geographic Scalability	-	◐	✓	✓	✓

→ **Space-time nesting as design choice**

Combine approaches deliberately to retain strengths and manage trade-offs

* LCFS = California Air Resources Board Low Carbon Fuel Standard

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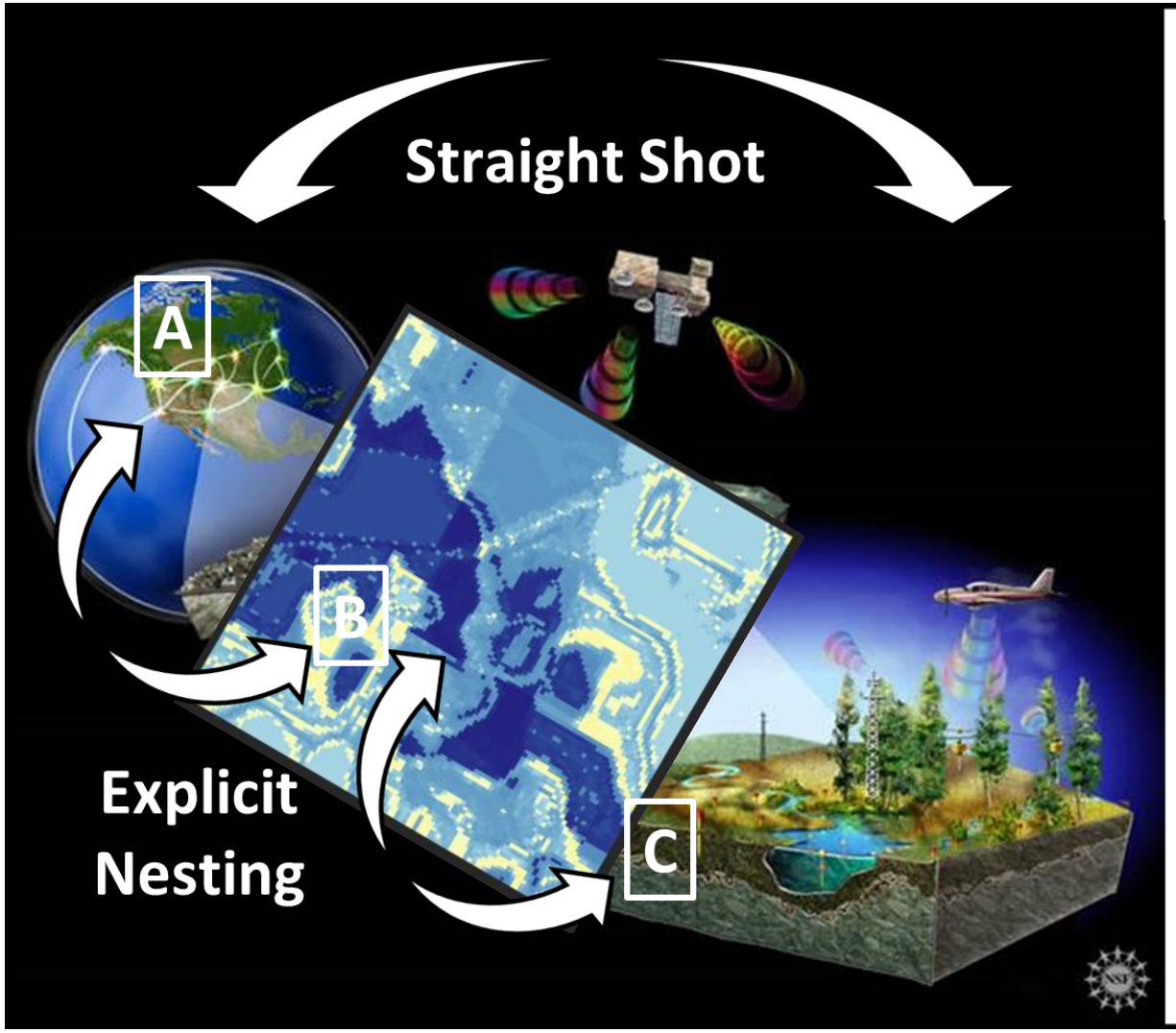
Workshop Motivation: Continuous Cross-Scale Resource Management

Simplicity

Transparency



Societal Relevance

A — Continental-to-global accountability and coordination of net-zero pledges, climate solutions, earth stewardship, e.g., UNFCCC nationally determined contributions.

B — Regional-to-continental information continuum for public awareness, policy-making from reliable stream of ground truth for satellites, climate solution testbed.

C — Local-to-regional practices across heat, water, carbon and pollution management sectors: heat islands, irrigation, emissions measurement, reporting and verification.

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